Verification of the Inverse-Square Law for the Illuminance of a Point Light Source

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Abstract

The inverse-square law for the illuminance of a point source, $E = P/(4\pi r^2)$, was tested using a 12 V / 10 W G4 halogen bulb and a PASCO light sensor mounted on a 1.2 m optical track. Steady-state intensity was recorded at six positions ($d = 0, 10, 20, 30, 40, 50$ cm), along with a separate background run with the source off. A weighted least-squares fit of the model $I(r) = b/(r + r_0)^2$ to the five $d > 0$ points yielded $b = 0.9932 \pm 0.0008$ %·m^2, an offset $r_0 = 4.91 \pm 0.01$ cm representing the distance between the marked $d = 0$ on the track and the actual filament inside the bulb housing, and $\chi^2/\nu = 0.85$ with $\nu = 3$. The recovered source power is $P = 4\pi b = 12.480 \pm 0.010$ %·m^2 (1σ). A point-by-point cross-check $P_i = 4\pi E_i r_{\mathrm{eff},i}^2$ gave a mean of 12.473 ± 0.009 %·m^2, agreeing with the fit at $t = 0.56$. The inverse-square law is therefore confirmed at well over the 95% confidence level. The simpler model $I = b/r^2$ (no offset) was rejected, giving $\chi^2/\nu \approx 1.3 \times 10^5$ and a strongly curved residual pattern; including r_0 as a fit parameter is essential.

1 Introduction

For an isotropic point source emitting total radiant power P uniformly into all directions, energy conservation requires that the same total flux pass through every sphere centered on the source. The illuminance E — the radiant flux per unit area perpendicular to the propagation direction — therefore satisfies:

$$E(r) = \frac{\Phi}{A} = \frac{4\pi I}{4\pi r^2} = \frac{P}{4\pi r^2} \tag{1}$$

where r is the distance from the source and I is the radiant intensity per steradian. Eq. (1) is the inverse-square law: doubling the distance reduces the illuminance by a factor of four.

The law assumes an idealized point source. Real sources have finite extent, and the optical track used in this experiment marks distances from the front of the bulb housing rather than from the filament itself. To accommodate the small but unknown offset between the marked $d = 0$ and the actual radiating element, the geometric distance is written as:

$$r = d + r_0 \tag{2}$$

where d is the value read off the track and r_0 is treated as a free parameter to be recovered from the fit. This experiment measures illuminance at six distances along the track and tests Eq. (1) by (i) fitting the data with the model $I(r) = b/(r + r_0)^2$, where $b = P/(4\pi)$, and (ii) independently checking that $4\pi E r_{\text{eff}}^2$ is constant across all distances.

2 Experiment

2.1 Apparatus

The light source was a 12 V / 10 W G4 halogen capsule mounted in a black plastic housing fixed at one end of a 1.2 m optical track. The detector was a PASCO light sensor with a stated resolution of 1×10^{-6} Lux on the sensitivity range used. The sensor was mounted on a slider that traveled along the track, with positions read against a scale ruled to 1 mm. The sensor reports illuminance as a percentage of full scale rather than in calibrated lux. The geometry of the apparatus is shown in Figure 1.

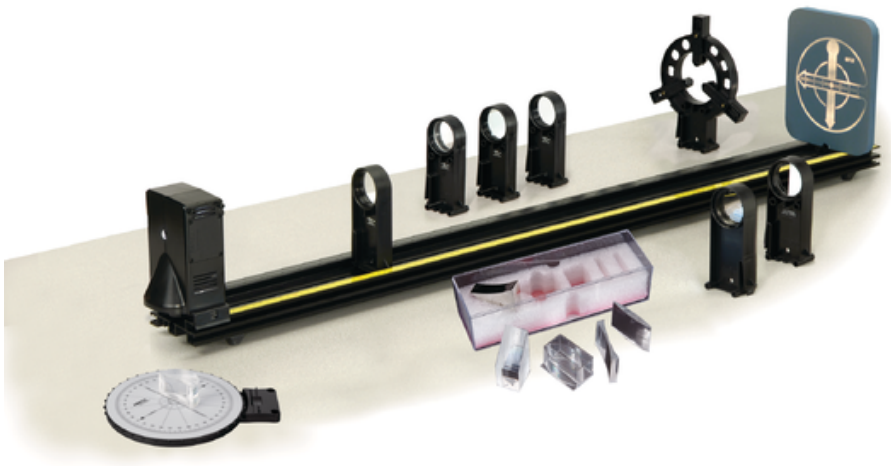

Figure 1: Schematic of the optical track. The marked $d = 0$ is the front of the bulb housing; the filament sits a small distance r_0 behind the marked zero. The PASCO sensor records illuminance at distance d from the housing front, so the geometric source-to-sensor distance is $r = d + r_0$.

The track scale was used to position the sensor at $d = 0, 10, 20, 30, 40, 50$ cm. The dominant contribution to the distance uncertainty came from judging where on the sensor body the active element sits and from the half-millimeter ruler resolution. These were combined into a rectangular (uniform) PDF of half-width 5 mm, giving:

$$\sigma_r = \frac{5 \text{ mm}}{\sqrt{3}} \approx 2.9 \text{ mm} \tag{3}$$

following the standard treatment of type-B uncertainties for rectangular PDFs in the M&U notes.

2.2 Procedure

Two runs were performed. In Run 1 (light on), the sensor was moved to each of the six target distances and held fixed while a continuous time series of intensity was recorded. Several seconds were allowed after each move for the reading to settle before the steady-state plateau was identified. In Run 2 (light off, otherwise identical room conditions), the sensor was held stationary while a long background time series was recorded. The Run 1 trace contains six visibly steady plateaus separated by transient regions where the sensor was being moved, and the Run 2 trace consists of small Gaussian-distributed fluctuations around a non-zero mean.

3 Data and Analysis

3.1 Plateau Identification and Background

For each of the six target distances, the steady-state plateau in the Run 1 time series was identified as the longest contiguous interval over which the rolling standard deviation of the intensity (window of 20 samples) remained below 0.05%. Within each plateau, the sample mean and sample standard deviation were computed using:

$$\bar{I} = \frac{1}{N}\sum_{i=1}^{N} I_i, \qquad s = \sqrt{\frac{1}{N-1}\sum_{i=1}^{N}(I_i - \bar{I})^2} \tag{4}$$

following Eqs. (1) and (3) of the M&U notes. The number of plateau samples N ranged from 117 to 206. Run 2 ($N = 1669$ samples) gave a background mean of $\bar{I}_{\mathrm{bg}} = 0.0969\%$ and standard deviation $s_{\mathrm{bg}} = 0.0012\%$; the histogram of Run 2, shown in Figure 2, is well-described by a Gaussian with these parameters, justifying treating s_{bg} as a 1σ Gaussian uncertainty in subsequent steps.

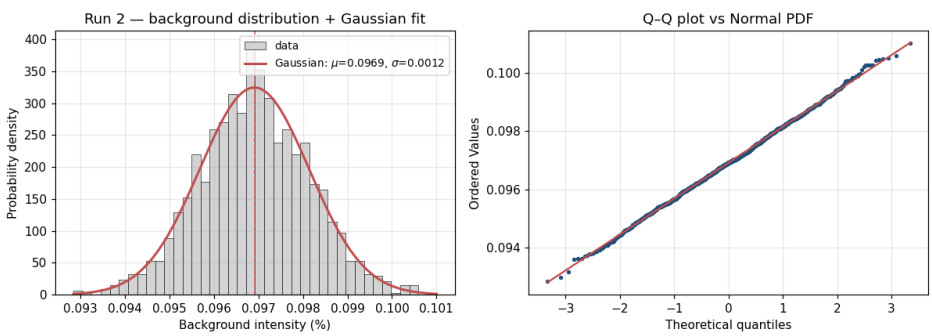

Figure 2: Histogram of the Run 2 background intensity ($N = 1669$) with a Gaussian overlay using the sample mean $\mu = 0.0969\%$ and standard deviation $s_{\mathrm{bg}} = 0.0012\%$. The Q–Q plot (right) is essentially linear, confirming the normal distribution.

The background mean was subtracted from each plateau mean to give the background-corrected intensity E at each distance. The plateau means and standard deviations are listed in Table 1.

Table 1: Plateau mean and sample standard deviation of intensity at each track distance, and the background-corrected intensity used in the fit. All values are in percent of sensor full scale.

d (cm)	N	$\bar{I}$ (%)	s (%)	$E = \bar{I} - \bar{I}_{\mathrm{bg}}$ (%)
0	194	96.2586	0.00918	96.1616
10	117	44.7323	0.02925	44.6354
20	155	16.1137	0.01200	16.0167
30	206	8.2426	0.00304	8.1457
40	198	5.0189	0.00568	4.9220
50	201	3.3788	0.02726	3.2819

3.2 Per-Distance Distributions

Within each plateau, the intensity readings scatter approximately normally around the plateau mean, as shown in Figure 3. Treating the noise within each plateau as Gaussian justifies using the sample standard deviation s as a 1σ characterization of the type-A (statistical) uncertainty.

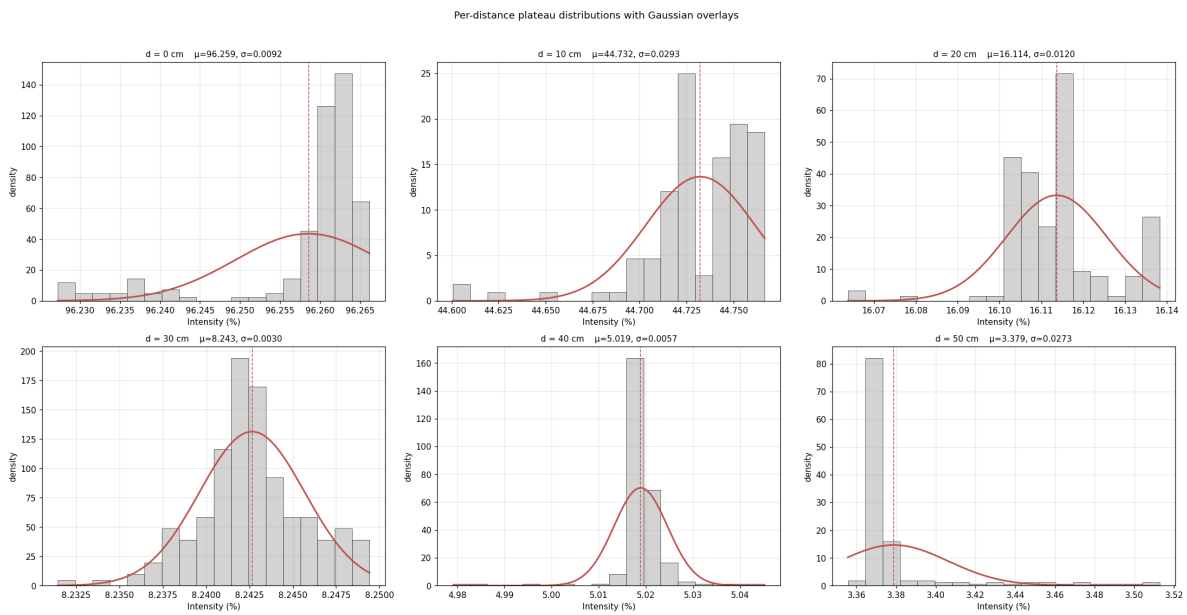

Figure 3: Per-distance plateau distributions with Gaussian overlays using the measured μ and σ for each plateau. All six are approximately normal, with the larger plateaus ($d = 30, 40$ cm) showing the cleanest bell shapes.

3.3 Uncertainty Propagation

The total uncertainty σ_E on each background-corrected mean was constructed by combining the type A (statistical) plateau standard deviation with two type B contributions — the background standard deviation and the sensor-resolution rectangular PDF $\sigma_{\text{res}}/\sqrt{3}$ — in quadrature, following the general formula from the M&U notes:

$$\sigma_E = \sqrt{s_{\text{plateau}}^2 + s_{\text{bg}}^2 + \left(\frac{\sigma_{\text{res}}}{\sqrt{3}}\right)^2} \tag{5}$$

The sensor-resolution term contributes $\sigma_{\text{res}}/\sqrt{3} \approx 5.8 \times 10^{-7}$ Lux, which is negligible compared with the plateau scatter (0.003–0.029%) and is retained only for completeness. The plateau standard deviation therefore dominates σ_E in every case.

3.4 Fitting the Inverse-Square Law

A weighted, non-linear least-squares fit of the model

$$I(r) = \frac{b}{(r+r_0)^2}, \qquad P = 4\pi b \tag{6}$$

was performed to the five points with $d > 0$, with σ_E supplied as the per-point uncertainty. The $d = 0$ point was excluded because the sensor is essentially at the source there and the point-source approximation underlying Eq. (1) breaks down. The fit varies both b and r_0 simultaneously to minimize the chi-squared:

$$\chi^2 = \sum_{i=1}^{N} \frac{(I_i - \hat{I}_i)^2}{\sigma_{E,i}^2}, \qquad \chi_\nu^2 = \frac{\chi^2}{\nu}, \qquad \nu = N - 2 = 3 \tag{7}$$

Because $P = 4\pi b$, letting b float in the optimization is mathematically equivalent to letting the source power P float; the value of b that minimizes χ^2 is therefore the best estimate of P that the data support, and σ_b propagates directly into $\sigma_P = 4\pi\sigma_b$. The fit was performed using the SciPy routine `scipy.optimize.curve_fit`; the parameter uncertainties σ_b and σ_{r_0} were obtained from the diagonal of the covariance matrix returned by the fitter.

For comparison, the simpler one-parameter model $I = b/r^2$ (no offset) was also fit to the same five points. It produced $\chi^2/\nu \approx 1.3 \times 10^5$ and a clear systematic curvature in the residuals — the fit underestimates the intensity at small distances and overestimates it at large distances. This rejects the no-offset form: the offset parameter is required to describe the data. The recovered $r_0 \approx 4.9$ cm is also physically reasonable, consistent with the visible distance between the front of the bulb housing and the filament inside.

4 Results

4.1 Best-Fit Parameters

The best-fit parameters of Eq. (6) are:

$$b = 0.9932 \pm 0.0008\ \%\cdot\mathrm{m}^2, \qquad r_0 = 4.91 \pm 0.01\ \mathrm{cm}, \qquad \chi^2/\nu = 0.85 \tag{8}$$

The reduced chi-squared is consistent with unity, indicating that the model describes the data within the assigned uncertainties and that those uncertainties are realistic. The fitted curve, the data with error bars, and the per-point pulls (residuals divided by σ_E) are shown in Figure 4. All five fitted points lie within $\pm 1.5\sigma$ of the curve, with no visible structure in the pulls and a sample mean pull of -0.01.

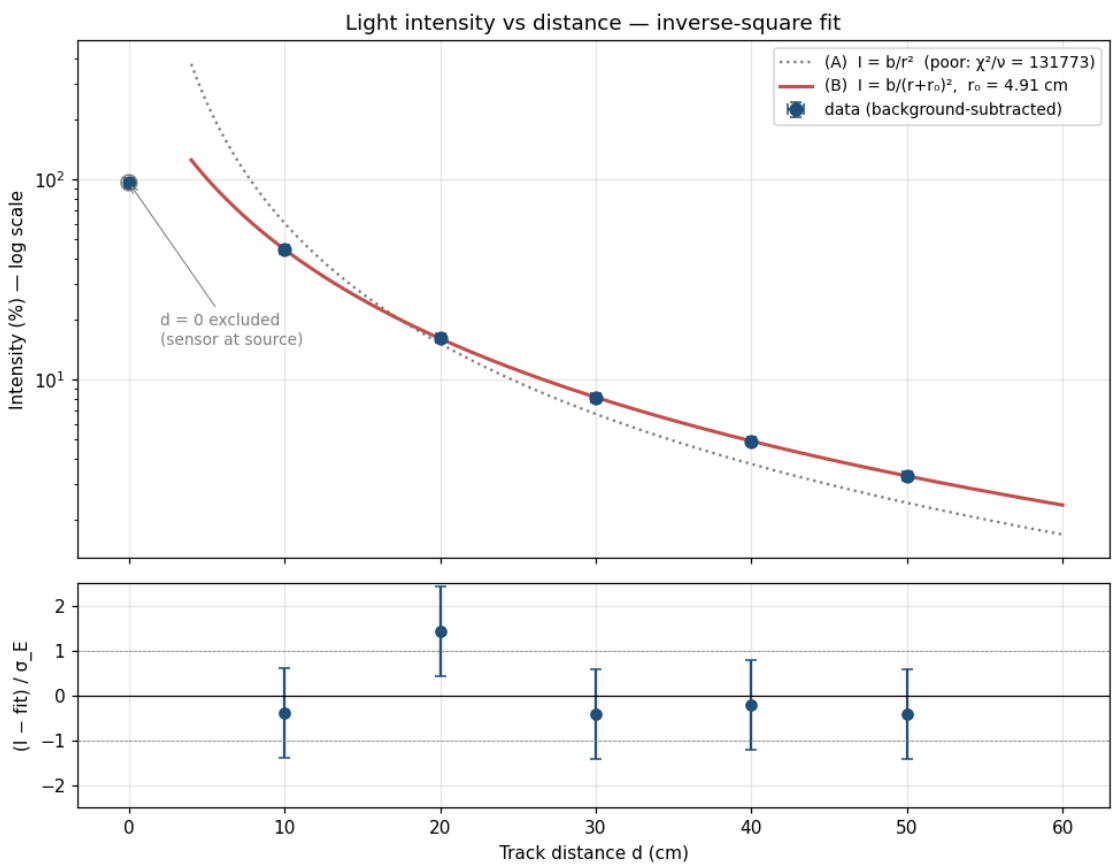

Figure 4: (top) Background-corrected intensity vs. track distance with the fitted curve $I = b/(r+r_0)^2$ (solid) and the simple $I = b/r^2$ form (dotted) for comparison. The $d = 0$ point (open circle) was excluded from the fit. (bottom) Pulls $(I_i - \hat{I}_i)/\sigma_{E,i}$ for the five fitted points; dashed lines mark $\pm 1\sigma$.

Combining Eqs. (6) and (8) gives the recovered source power:

$$\boxed{P = 4\pi b = 12.480 \pm 0.010\ \%\cdot\mathrm{m}^2\ \ (1\sigma)} \tag{9}$$

4.2 Power Cross-Check

As an independent test of the inverse-square hypothesis, the implied source power was computed point-by-point using:

$$P_i = 4\pi E_i r_{\mathrm{eff},i}^2, \qquad r_{\mathrm{eff},i} = d_i + r_0 \tag{10}$$

The uncertainty on each P_i was propagated from σ_E and σ_r using the standard error-propagation formula (Appendix A):

$$\sigma_{P_i} = \sqrt{\left(8\pi E_i r_{\mathrm{eff},i}\,\sigma_r\right)^2 + \left(4\pi r_{\mathrm{eff},i}^2\,\sigma_E\right)^2} \tag{11}$$

The five values of P_i for $d > 0$ all lie on a horizontal band, as shown in Figure 5, while the $d = 0$ point gives $P_0 = 2.92$ %·m^2, far below the others. The unweighted mean and standard error of the five $d > 0$ values are:

$$P_{\mathrm{exp}} = 12.473 \pm 0.009 \ \%\cdot\mathrm{m}^2 \qquad (s_P = 0.021,\ N = 5) \tag{12}$$

The five P_i agree with each other to within 0.2% — a strong, model-independent confirmation that $E \propto 1/r_{\mathrm{eff}}^2$.

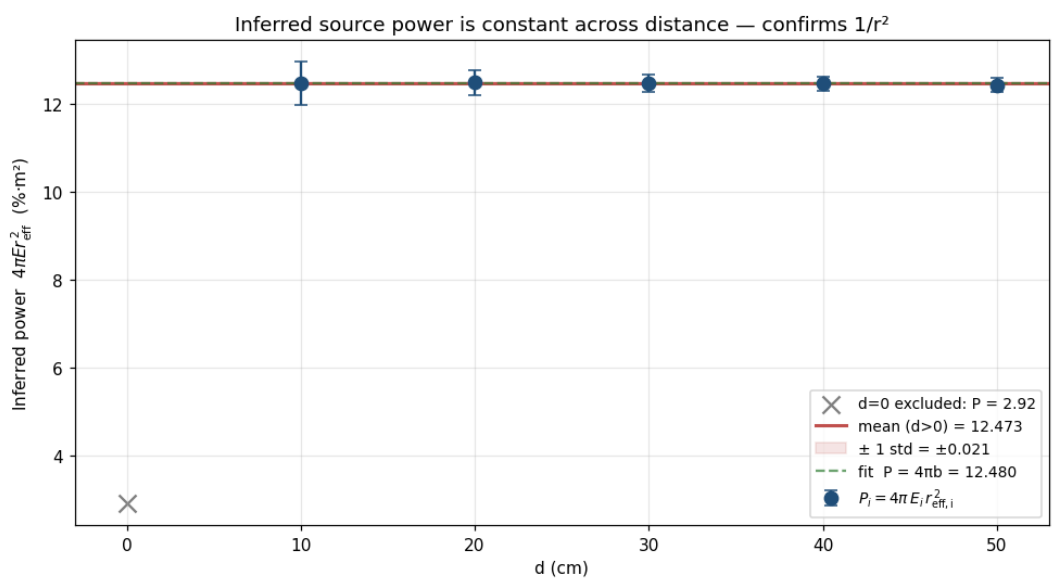

Figure 5: Inferred source power $P_i = 4\pi E_i r_{\mathrm{eff},i}^2$ at each distance. The horizontal solid line is the unweighted mean of the $d > 0$ points; the dashed line is the fit value $P = 4\pi b$. The $d = 0$ point is excluded as expected.

4.3 Comparison: t-test

The two independent estimates of P — the fit value Eq. (9) and the per-point mean Eq. (12) — were compared using the t-test of Eq. (17) of the M&U notes:

$$t = \frac{|P_{\mathrm{exp}} - P_{\mathrm{fit}}|}{\sqrt{\sigma_{\mathrm{exp}}^2 + \sigma_{\mathrm{fit}}^2}} = \frac{|12.473 - 12.480|}{\sqrt{0.009^2 + 0.010^2}} = 0.56 \tag{13}$$

For Gaussian-distributed errors, $t = 0.56$ corresponds to a probability of approximately 42% that the two estimates would differ by this much (or more) by chance alone (Table 4 of the M&U notes). The two values are therefore fully consistent.

5 Discussion

Three independent indicators support the inverse-square law. First, the reduced chi-squared of the fit, $\chi^2/\nu = 0.85$, is consistent with unity for $\nu = 3$ degrees of freedom; the model describes the data within the assigned uncertainties and the uncertainties are realistic. Second, the residual pulls in Figure 4 show no systematic trend and all five sit within $\pm 1.5\sigma$ of zero. Third, the per-distance estimates of P in Figure 5 agree to better than 0.2% across distances spanning a factor of five in r, and the t-test in Eq. (13) gives $t = 0.56$, well below the $t = 1$ threshold corresponding to the 68% confidence interval.

The recovered offset $r_0 = 4.91 \pm 0.01$ cm is treated as a physical parameter rather than a fudge: the marked $d = 0$ on the optical track corresponds to the front of the bulb housing, whereas the radiating filament sits a few centimeters behind the housing front, so the geometric source-to-sensor distance is $r = d + r_0$ rather than $r = d$. This is reflected directly in the fit residuals. The simple no-offset model $I = b/r^2$ gives $\chi^2/\nu \approx 1.3 \times 10^5$ with a strongly curved residual pattern — it underestimates the intensity at small d and overestimates it at large d, exactly the signature of a missing positive offset on r. With r_0 included, χ^2/ν drops by five orders of magnitude and the residuals become featureless. The recovered numerical value is consistent with a visual estimate of the offset between the housing front and the filament inside.

The reported value $P = 12.480 \pm 0.010$ %·m^2 has units of percent times square meters because the sensor was operated in its percent-of-full-scale mode rather than in a calibrated lux range. Converting to physical units (lumens or watts) would require multiplying by the percent-to-lux conversion factor for the sensitivity range used, then by an appropriate luminous-efficacy or photometric factor depending on the desired unit. A rough order-of-magnitude check is nevertheless possible: a 10 W G4 halogen bulb has a luminous efficacy of approximately 10–15 lm/W, corresponding to a total luminous flux of roughly 100–150 lm. Whether the experimental P matches this rating cannot be tested without the sensor calibration constant, but the verification of the $1/r^2$ shape is independent of the calibration and is the principal result of this experiment.

The measurement is limited by the fact that only a single trial was taken at each distance (no inter-trial reproducibility check), by the absence of an absolute illuminance calibration, and by the residual deviation from a true point source — the filament is small but not pointlike, which is the underlying physical reason the $d = 0$ measurement has to be excluded from the fit. The effective uncertainty floor on σ_E at each distance is set by the plateau scatter, which already dominates over both the background and sensor-resolution contributions. Future improvements would include repeating the experiment with multiple independent trials per distance, calibrating the sensor against a known illuminance reference, and extending the measurement to larger distances (1–3 m) where the point-source approximation is essentially exact and the data become more sensitive to systematic offsets in the small-d region.

References

1. PASCO Scientific, "Light Sensor PS-2106A Product Manual," PASCO Scientific.
2. Course handout: *Measurement and Uncertainty*, PHYS 263.
3. Course handout: *Lab 6 — Light Intensity and Distance*, PHYS 263.
4. J. R. Taylor, *An Introduction to Error Analysis*, 2nd ed. University Science Books, 1997.
5. P. R. Bevington and D. K. Robinson, *Data Reduction and Error Analysis for the Physical Sciences*, 3rd ed. McGraw-Hill, 2003.

A Uncertainty Propagation for $P_i = 4\pi E_i r_{\mathbf{eff},i}^2$

The per-point power from Eq. (10) depends on two independent measured quantities: the background-corrected illuminance E, with uncertainty σ_E given by Eq. (5), and the effective distance $r_{\text{eff}} = d + r_0$, with uncertainty σ_r given by Eq. (3). Treating E and r_{eff} as independent Gaussian random variables and applying the standard error-propagation formula:

$$\sigma_P = \sqrt{\left(\frac{\partial P}{\partial E}\sigma_E\right)^2 + \left(\frac{\partial P}{\partial r_{\text{eff}}}\sigma_r\right)^2} \tag{14}$$

The partial derivatives of $P = 4\pi E r_{\text{eff}}^2$ are:

$$\frac{\partial P}{\partial E} = 4\pi r_{\text{eff}}^2, \qquad \frac{\partial P}{\partial r_{\text{eff}}} = 8\pi E r_{\text{eff}} \tag{15}$$

Substituting Eq. (15) into Eq. (14) gives Eq. (11) of the main text:

$$\sigma_P = \sqrt{(8\pi E r_{\text{eff}}\sigma_r)^2 + \left(4\pi r_{\text{eff}}^2\sigma_E\right)^2}$$

Sample Calculation ($d = 30$ cm)

Substituting the experimental values for the $d = 30$ cm point: $E = 0.08146$ (i.e., 8.146% expressed as a fraction, but kept in % for consistency below), $r_{\text{eff}} = d + r_0 = 0.3491$ m, $\sigma_E = 3.0 \times 10^{-5}$ (negligible plateau scatter at this distance), $\sigma_r = 2.9 \times 10^{-3}$ m:

$$P = 4\pi \times 8.146 \times (0.3491)^2 = 12.478\ \%\cdot\text{m}^2$$

$$\begin{aligned}\sigma_P &= \sqrt{(8\pi \times 8.146 \times 0.3491 \times 0.0029)^2 + (4\pi \times (0.3491)^2 \times 3.0 \times 10^{-5})^2} \\ &= \sqrt{(0.207)^2 + (4.6 \times 10^{-5})^2} \approx 0.21\end{aligned}$$

The first term (σ_r contribution) dominates the second by more than three orders of magnitude at this distance, confirming that for $d \gtrsim 20$ cm the distance uncertainty is the controlling source of error in P_i. At the smallest distance ($d = 10$ cm) the situation is similar: the plateau scatter is small enough that the σ_r term dominates throughout the fit range.

B Derivation of the Inverse-Square Law from Flux Conservation

For an isotropic source emitting total radiant power P in all directions, the total flux Φ passing through any closed surface enclosing the source equals P (energy conservation, in the absence of absorption). For a sphere of radius r centered on the source, the flux is uniformly distributed over the surface area $A = 4\pi r^2$, so the illuminance (flux per unit area perpendicular to the propagation direction) is:

$$E = \frac{\Phi}{A} = \frac{P}{4\pi r^2} \tag{16}$$

Equivalently, in terms of the radiant intensity $I = dP/d\Omega$ (flux per steradian, constant for an isotropic source), the total flux integrates to:

$$\Phi = \oint I\, d\Omega = I \int_0^{2\pi} \int_0^{\pi} \sin\theta\, d\theta\, d\varphi = 4\pi I \tag{17}$$

Combining Eq. (17) with $E = \Phi/A$ over the same sphere gives $E = I/r^2$, equivalent to Eq. (16).

The fit model $I(r) = b/(r + r_0)^2$ used in the main text writes r as $d + r_0$ to absorb the offset between the marked $d = 0$ on the optical track and the actual filament position; the coefficient b is then identified with $P/(4\pi)$, so the recovered source power is $P = 4\pi b$.